

FINANCE AND MARKET TERMS

StockPulse Glossary

How StockPulse calculates costs, forecast edge, portfolio weights, and order risk. Definitions match the live dashboard and API - not a textbook. Nothing here is investment advice.

Date: 24 August 2026

Product: StockPulse (Kronos monorepo)

Local UI: <http://localhost:5173>

LAN UI: <http://192.168.86.197:5173>

Not investment advice. Forecasts are research outputs. Broker acceptance does not guarantee execution. Prefer paper mode until you understand the risk controls.

1. How to use this glossary

Terms are grouped by product surface. Formulas mirror `backend/app/services/research.py` (forecast annotations) and `backend/app/services/risk.py` (order preview). Defaults come from `backend/app/config.py` and can be overridden in `backend/.env`. Forecasts never auto-trade.

2. Market data

OHLCV

Open, High, Low, Close, Volume for one bar (candle). The chart and Kronos context window are built from these fields.

Bar / candle

One OHLCV sample over a fixed interval. Short horizon uses 5-minute bars; long horizon uses daily bars.

Session / RTH

Regular trading hours for US equities (roughly 09:30-16:00 America/New_York). Forecast timestamps skip weekends and stay inside RTH for intraday bars.

Pre-market / after-hours

Extended sessions outside RTH. The market clock badge reflects Alpaca session state. Intraday forecast synthesis still anchors to RTH open/close rules.

Quote

Latest tradeable snapshot for a symbol: last price, day change, OHLC, volume, and optional fundamentals (market cap, P/E, EPS, dividend yield).

ADV (average daily volume)

Typical share volume from the quote snapshot. Risk checks require $ADV \geq RISK_MIN_ADV_SHARES$ (default 200,000) before allowing a buy. Cost helpers also use dollar $ADV = \text{mean}(\text{volume} * \text{close})$ when estimating slippage.

Bid / ask / mid

Best buy and sell quotes. Quoted spread in bps = $(\text{ask} - \text{bid}) / \text{mid} * 10,000$ when both sides exist. Used for live risk checks on buys.

3. Forecast engines and horizons

Kronos mode

Chart engine toggle (engine=kronos). Single NeoQuasar Kronos path (or labeled baseline-drift if Torch cannot load). Movers scans always use this path.

Forecast mode

Chart engine toggle (engine=ensemble). Weighted blend of enabled models from forecasting/ (persistence, Kronos, Chronos, TimesFM, ...). If only persistence survives, StockPulse falls back to the Kronos/baseline path and marks the response as ensemble_degraded.

Short horizon

Preset on 5-minute bars. Default context 256, default horizon 12 bars; UI can override horizon to 1 / 10 / 20 / 30 / 60.

Long horizon

Preset on daily bars. Default context 256, default horizon 20 trading days; UI can override to 1 / 10 / 20 / 30 / 60.

Context

Historical bars the model may see. StockPulse caps context at min(requested, KRONOS_MAX_CONTEXT, 512) for Kronos-small.

Forecast path

Sequence of predicted closes (Kronos usually returns full OHLCV) plotted after the last historical candle.

Path segments

UI breakdown of the forecast into up / down / flat stretches for decision context.

Direction noise band

Trend is flat when $|\text{forecast_change}| < 0.002$ (0.2%). Same band is used when scoring walk-forward direction hits.

Baseline-drift

Fallback when Kronos/PyTorch cannot load: last close grown by the mean of the last ~20 bar returns, each step clamped to +/- 2%. Labeled in model.id so it is not mistaken for a neural forecast.

4. Cost and edge calculations

Round-trip cost (research.py)

For each forecast annotation StockPulse estimates friction from the same OHLCV window used as context (default notional assumption \$10,000):

```
range_i      = (high_i - low_i) / close_i
spread_bps   = clamp(mean(range) * 10000, floor=2, cap=80)
realized     = stdev(close returns over last ~21 bars)
dollar_ADV   = mean(volume_i * close_i)
particip.    = min(1, notional / dollar_ADV)  # else 0.05
slip_bps     = clamp(10000 * realized * sqrt(particip.) * 0.5, 0.5, 40)
fee_bps      = 0.5
round_trip   = 2 * (spread_bps + slip_bps) + fee_bps
```

If fewer than 5 valid bars exist, a conservative default is used: $2 * 2\text{bp spread floor} + 2\text{bp slip} + 0.5\text{bp fee}$.

Basis point (bp / bps)

One hundredth of one percent. $100 \text{ bps} = 1\%$. Dividing bps by 10,000 converts to a decimal return haircut.

Gross and net forecast change

```
gross = last_predicted_close / last_historical_close - 1
net   = sign(gross) * (|gross| - round_trip_bps/10000)
      = -round_trip_bps/10000   when gross == 0
```

Net change is what the chart meta line and movers ranking use after the cost haircut. A positive net means the predicted move still exceeds estimated friction.

Quoted spread (risk path)

Separate from the bar-range spread proxy: $(\text{ask} - \text{bid}) / \text{mid} * 10,000$ from the live snapshot. Used only in order risk checks.

5. Regime, walk-forward, and edge reliability

Regime label (classify_regime)

Needs ≥ 21 closes. $\text{realized_vol} = \text{stdev of last 20 bar returns}$. Compare short vol to longer (up to 60) vol; trend compares 20-bar vs 60-bar mean price ($\pm 0.5\%$ band = sideways). Label form: $\{\text{low|normal|high}\}_{\text{vol}}_{\{\text{up|down|sideways}\}}$.

Walk-forward evaluation

Kronos mode (not ensemble) may run purged walk-forward folds (KRONOS_EVAL_FOLDS, default 3; context KRONOS_EVAL_CONTEXT, default 32). Eval horizon is $\min(\text{path_horizon}, 20)$. For each fold:

```
signal at bar t using only bars [t-context, t)
fill at next bar open (or close if open missing)
exit at close of bar t+horizon-1
traded_return = predicted_dir * realized_return   # 0 if flat signal
net_fold = traded_return - round_trip_bps/10000   # 0 if no trade
```

Hit rate

Share of folds where predicted_dir matches realized_dir (both non-flat).

Mean net return / IC

Average net_fold across folds; IC is Spearman rank correlation of predicted gross vs realized when enough folds exist.

Edge reliable

True only if folds ≥ 2 , mean_net_return > 0 , hit_rate ≥ 0.5 , and (when the current regime bucket has ≥ 2 folds) that bucket's mean_net_return is also > 0 . Dims the Investors sentiment badge when false.

Live OOS journal

StockPulse appends forecast_change snapshots to an in-memory / optional file journal and scores them later against realized bars (live_oos in evaluation).

6. Portfolio and open positions

Equity

Broker-reported account equity for the selected paper/live mode. Shown in the account card and Open positions summary.

Buying power / cash

Broker capacity for new buys and settled cash. Buys that exceed buying power are blocked at preview.

Market value

Alpaca position market_value for the holding (signed; UI sorts by absolute size).

% of equity (Open positions)

portfolio_weight = position.market_value / account.equity. Displayed as a percent (for example 1.20%). Shows em dash if equity is missing or zero.

Unrealized P/L

Broker unrealized dollar P/L and optional percent vs average entry. Separate from portfolio weight.

Gross exposure

Sum of abs(market_value) across open positions. Used in risk: after a buy, (gross_exposure + order_cost) / equity must stay $\leq$ RISK_MAX_GROSS_PCT.

Daily P/L percent

daily_pnl_pct = equity / last_equity - 1. If this is at or below -RISK_MAX_DAILY_LOSS_PCT, new buys are halted (new_buys_halted).

7. Order sizing and review preview

Estimated notional

```
equity: notional field, else qty * (limit|stop|last|ask)
option: same, then * 100 (contract multiplier)
```

position_pct (preview)

For buys: $(\text{current_position_value} + \text{estimated_cost}) / \text{equity}$. For sells: $\text{current_position_value} / \text{equity}$. Shown in the review dialog as "X% of equity".

Review-before-send

Every equity/option order opens a confirmation dialog with estimated cost, position %, spread bps, daily P/L %, and warnings. Nothing routes from the forecast engine automatically.

Large market-order warning

Soft warning (does not hard-block) when type=market and estimated cost $\geq$ \$5,000.

8. Risk portfolio controls (defaults)

Hard limits live in `check_order_risk`. Breaches raise `ValueError` and become 422/preview failures. Defaults (override via env):

Setting	Default	Rule
RISK_MAX_POSITION_PCT	10%	Buy: $(\text{pos} + \text{cost}) / \text{equity} \leq \text{limit}$
RISK_MAX_OPTION_DEBIT_PCT	5%	Same for option buys (debit)
RISK_MAX_GROSS_PCT	150%	$(\text{gross} + \text{cost}) / \text{equity} \leq \text{limit}$
RISK_MAX_DAILY_LOSS_PCT	3%	Buy blocked if daily P/L $\leq$ -limit
RISK_MAX_SPREAD_BPS	80	Buy blocked if quoted spread $>$ limit
RISK_MIN_ADV_SHARES	200,000	Buy blocked if ADV below floor
ALLOW_SHORT_SELLING	false	Equity sell qty/notional $\leq$ long
ALLOW_UNCOVERED_OPTIONS	false	Blocks sell_to_open by default

Sell-to-close / buy-to-close

Option closes cannot exceed the held long/short contract quantity.

Live confirmation

Live mode requires `ALLOW_LIVE_TRADING` on the server plus typing `LIVE` (and matching `live_confirmation_token` on the order).

9. Sentiment badges

Public sentiment

News-derived bullish / bearish / neutral (Finnhub-oriented).

Investors sentiment

Mapped from forecast trend direction (up->bullish, down->bearish, flat->neutral). Dimmed when `edge_reliable` is false.

10. Movers scan

Movers ranking

Background Kronos forecasts over a liquid universe (Alpaca actives/movers + fallback list). Ranked by absolute net_forecast_change after the cost haircut. UI shows top predicted gainers and losers (cap 50).

Scan rate limit

Separate throttle from per-ticker forecasts so a universe scan does not starve the chart (defaults: ~10 scan req/min vs ~30 ticker forecasts/min).

Evaluate=False on scan

Movers forecasts skip walk-forward folds for speed; chart Kronos mode still may run OOS folds.

11. Trading account vocabulary

Paper / live

Simulated vs real Alpaca account. Default UI mode is paper.

Notional vs qty

Dollar size vs share/contract count. Exactly one is required for equity orders.

Market / limit / stop

Market seeks immediate fill; limit at your price or better; stop/stop-limit need stop_price (and limit_price for stop-limit).

Time in force

Day cancels at session end; GTC may persist (broker rules apply).

12. Options (single-leg)

Contract symbol

OCC-style option identifier (underlying + expiry + call/put + strike).

Buy to open / sell to close

Open a long option / close an existing long.

Sell to open / buy to close

Open a short option / cover it. Sell-to-open requires ALLOW_UNCOVERED_OPTIONS=true.

13. Research extras

Visible mainly in forecasting/ and kronos_backtest/, and sometimes in logs.

Look-ahead bias

Using information unknowable at decision time. The production backtester refuses predictors that see future bars.

RankIC

Spearman correlation of predicted scores vs realized outcomes across names.

Ensemble disagreement

Cross-model std of final closes / path. Wide disagreement is a low-confidence signal even if the blend looks calm.

Weighted average ensemble

Default Forecast-mode blend: $\text{sum}(w_i * \text{close}_i) / \text{sum}(w_i)$ using weights from forecasting/config/models.yaml.